Existential Factors vs. Academic Performance

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Abstract

How much do existential factors play a role in a student's academic performance? The answer to this question can solve many problems regarding the common misconception that being smart is genetic or natural. As the research performed disproves this idea, it shows that students can all have closer to equal opportunity than society makes it out to be. We attained a data set that contained information about 2,392 high school students with details about their demographics, study habits, opportunities, extracurriculars, background and more. Using this data set, we trained multiple machine learning models including lasso regression, ridge regression, and others to recognize patterns throughout the student's information, checking for correlation with each of the factors and their grade point average (GPA). The results confirmed that some of the factors in the data set had impacts on the students' GPA. While measuring a variety of different combinations of factors, we found the r^2 approximately 93% to their GPA with a specific set. The major conclusion in relation to impact on existential factors is that these factors do impact students' academic performance. The ones that had major correlation were study time (weekly), parental support, absences, tutoring, and parental education.

Introduction

How much do confounding factors play a role in a student's academic performance? In regards to "confounding" factors, this means any factors that might be impacting the students that might not be considered. This research question holds substantial significance in the matter of how academically exceptional a student is able to be. Some students hold themselves accountable and have strict standards for themselves in their academics while others think that it may be out of their control and do not take initiative. This internal and external locus of control can be conflicting when trying to identify the truth about the question, is it in the student's control. The reason that this question holds much significance is because the answer can prove to students that they have power and influence over their academic performance. To start off this project, we decided to find a data set that would accurately represent the general population of students. Our next step would be to analyze the data by checking each factor represented throughout the data set. The features that were easy to eliminate would be eliminated prior to any testing and the ones that could potentially have correlation would remain on the list to be tested. Our next task would be to implement machine learning by training the program with the data in order for it to recognize correlation between the x and y axis. We planned to use all sorts of models like linear regression, random forest regression, ridge regression, and others to check for the most accuracy.

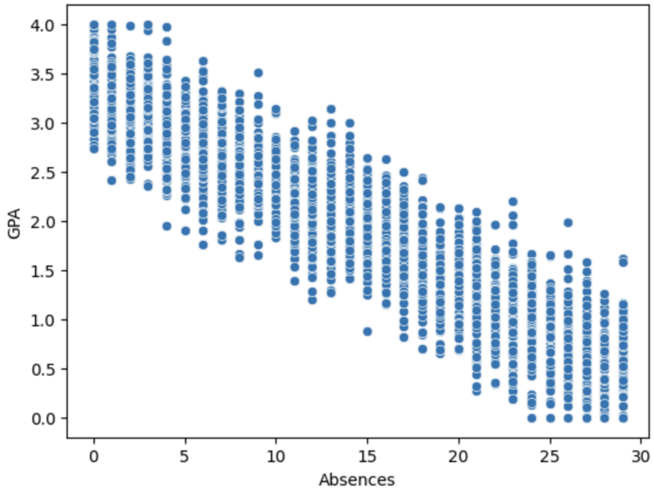
Background

In another article, *Study Habits of Highly Effective Medical Students*, written by Khalid Abdulrahman, they had a similar research problem in which they solved a different way. The researchers conducted a cross-sectional study and utilized standardized questionnaires to gather information about various study habits. The 675 students that were given the questionnaire were all medical students in Saudi Arabia. The researchers employed and ensured consistency by doing this. They achieved results which proved significant correlation between certain study habits and academic performance among these medical students. Some of which included effective time management, eliminating interruptions, setting goals to prioritize important activities, studying three to four hours daily, studying alone, learning from multiple sources, engaging in peer teaching, reviewing main lecture slides, combining lecture slides with notes, and maintaining motivation. Some limitations included that students may have learned differently and certain study habits would not be as effective on different types of students which is similar with our research project as well. Another factor was that the information was self-reported meaning that dishonesty could've impacted the data.

Dataset

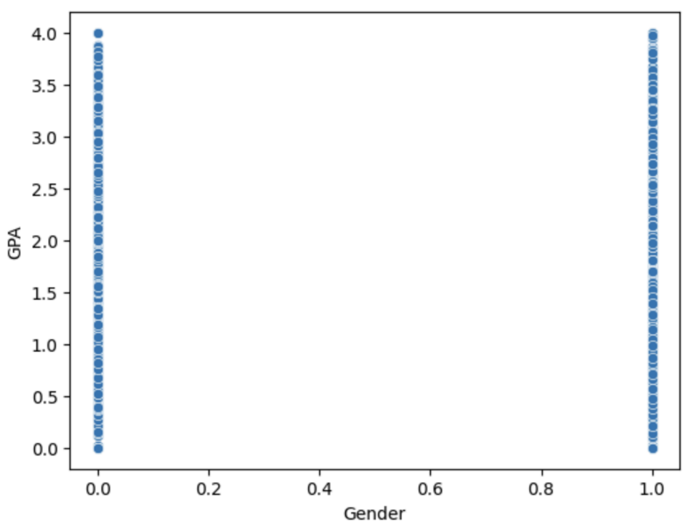
The dataset we used in our research project was created by Rabie El Kharoua called *Students Performance Dataset*. The dataset contains information on about 2,392 high school students and it highlights their demographics, study habits, parental involvement, extracurricular activities, and academic performance, all in numerical data format. The dependent variable that was being measured was their Grade Class, or their GPA

Some specific models showed direct correlation independently when measured against the GPA. One of which included Absences compared to GPA. We noticed that as students were absent more, their grade point averages tended to decrease.



As shown on this graph comparing absences to GPA, the more absences students attained, the lower their GPA tended to be in a broad scale.

Other models didn't show any correlation between the independent variable and students' grade point average. One of which included gender which was expected. As shown in the graph below, there are students with grade point averages that range from 0 all the way to 4 with no correlation to gender.



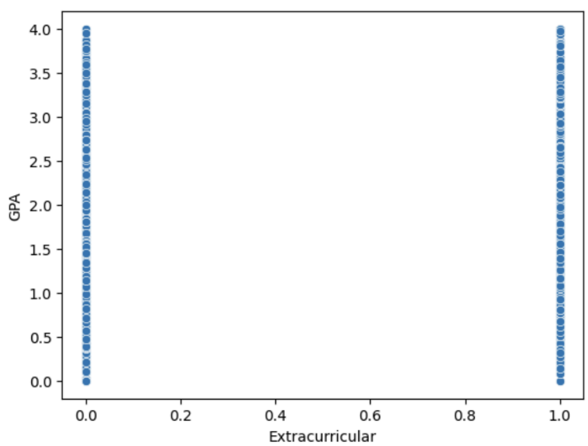
On this graph, 0 represents a woman and 1 represents a man. As clearly depicted, the GPA ranges from 0 all the way to 4, regardless of the gender difference.

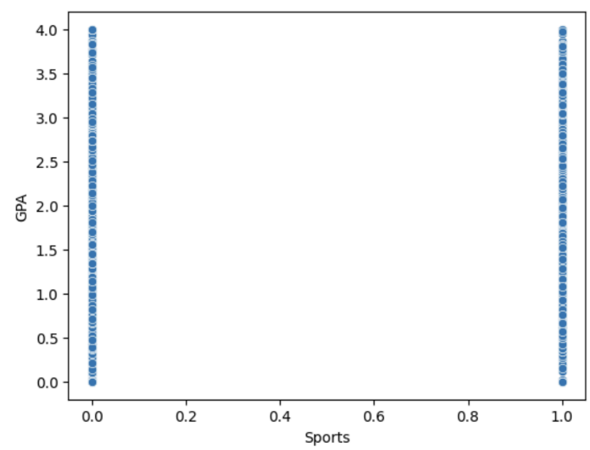
Methodology/Models

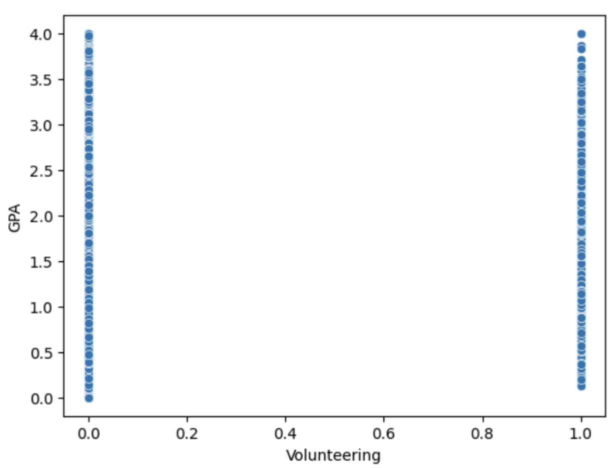
First, we took apart the data and looked for patterns with individual independent variables and compared them to the students' grade point average (GPA). We compared them to the grade point average because that is what measures their academic success. We looked at certain factors that we suspected might have a connection to their GPA. These included parental education, study time weekly, absences, tutoring, parental support, involvement in extracurricular activities, sports, and volunteering so we cut the rest from the data.

To do this, we created a new variable, data_mod, and assigned it to the data so we didn't affect the real data. Then, we dropped the factors that didn't seem to have any importance using data_mod = data_mod.drop([‘StudentID’, ‘Age’, ‘Gender’, ‘Music’], axis = 1).

To check for relationships between these groups and grade point averages, we plotted the data using scatter plots. After doing this, we removed some factors from the final experimental group that didn't show any correlation. These included involvement in extracurriculars, sports, and volunteering.







Shown on these graphs, students' GPA ranged heavily from 0 through 4 whether they participated in the displayed extracurricular activities or not.
This left us with the final set of factors that impacted the students' GPA which were study time weekly, parental support,

absences, tutoring, and parental education.

We trained the machine learning model with the x and y values using the x_train and y_train function. Using this, the model learned the patterns and relationships of the data during this phase. The x_test and y_test functions were also used after that to check the prediction accuracy.

Ridge regression model is a type of linear regression that adds a penalty to prevent overfitting which leads to more accurate and reliable predictions. These penalties are called L2 regularization which keeps the coefficients (weights) smaller and helps the model generalize better.

The Random Forest Regressor is a model which uses many decision trees to make predictions on the data. It builds multiple trees, each trained on a random subset of the data, and averages it all out to make accurate and stable predictions. This technique helps to reduce overfitting as well and improves generalization.

Lasso regression is another type of linear regression which adds a penalty to prevent overfitting, but instead of Ridge regression's L2 regularization, it uses L1 regularization. This penalty forces some coefficients to become exactly zero, effectively selecting only the most important features for the model. The strength can be controlled through a parameter.

After this, we used a ridge regression model to insert the x_test and y_test data. Then the y_pred variable was assigned to be the prediction of the model based on the training done earlier. We calculated the mean absolute error, mean squared error, and the correlation percentage through the mae, mse, and r^2 functions.

The code segments we used to set up the ridge regression model were; ridge = Ridge(alpha=1.0), ridge.fit(x_train, y_train), y_pred = ridge.predict(x_test). The formulas we used to calculate the mae, mse, and r^2 were as followed; mae = mean_absolute_error(y_test, y_pred), mse = mean_squared_error(y_test, y_pred), r2 = r2_score(y_test, y_pred).

Results

While trying to find the best model to represent the data, we began with Random Forest Regressor which gave mediocre results. As we attempted the default hyperparameter of n_estimators which was 100. This function specifies the number of trees in the forest model. The results for this test concluded in the mean absolute error to be 0.308, the mse to be 0.152 and the r^2 was 0.806. As we experimented further with the n_estimators hyperparameter, we found that as we increased the number, the results would improve and as we decreased the number, the results would decline, but the results were very minimal overall.

Then, we moved onto a Lasso Regression model, experimenting with the alpha parameter. This controls the amount of shrinkage that is applied to the equation. The default number for the parameter was 1.0 and the results for this concluded to the mean absolute error being 0.296, the mean squared error being 0.137, and the r^2 being 0.834. As we increased the alpha, the results declined. As we decreased the alpha to 0.1, the results were noticeably better than the default trial. The mean absolute error was 0.255, the mean squared error was 0.103, and the r^2 was 0.875. We also tried to decrease the alpha to 0.0 in which the results outperformed the previous attempt. The results were the mean absolute error being 0.245, the mean squared error being 0.096, and the r^2 being 0.884.

Although these results were adequate, we attempted another model, the Ride Regression. In this model, we also chose to experiment with the alpha hyperparameter. The default value was 1.0 so that is what we started with. The results came out to the mean absolute error being 0.193, the mean squared error being 0.058, and the r^2 being 0.934. These results significantly exceeded the previous results with the other models. As we increased and decreased the alpha, the results declined slightly.

Random Forest Regressor			
Change	n_estimators = 100 (default)		n_estimators = 200
MAE	0.3084859263		0.3247002998
MSE	0.1523850372		0.1655257615
R^2	0.8063859367		0.801108793
Lasso Regression			
Change	alpha=0.1	alpha=1.0	alpha=0.0
MAE	0.2552958735	0.2962956401	**0.2452244921**
MSE	0.1031028594	0.1373830572	**0.09607192559**
R^2	0.8753820573	0.8342947653	**0.8845626139**
Ridge Regression			
Change	alpha = 1.0	alpha = 0.1	alpha = 10.0
MAE	**0.1935095555**	0.2452216646	0.2449633073
MSE	**0.05862755909**	0.09607036568	0.09592732579
R^2	**0.9340613947**	0.8845644883	0.8847363611

In the figure above, the table with all the results are shown, with the results highlighted in green being the most significant.

Discussion

The findings from this confirm that there is a strong correlation between a student's grade point average and the factors measured; study time weekly, parental support, absences, tutoring, and parental education. Although an earlier limitation of strategies not applying to everyone the same way was brought up, it is highly probable that at least one of these factors can help a student improve their GPA. While some of these may seem obvious and some are unattainable after a certain point like parental education, partaking in the rest of these can greatly improve a student's chances at attaining their goal grade point average. Something that teachers can do to influence that is educating their students on these factors and how they play a huge role in their academic success. The more studying, education your parents attained, parental support, and tutoring a student has access to will dramatically influence their GPA. On the other hand, the more absences a student has, the more their GPA will decrease.

A limitation that we recognized was that the Ridge Regression model can only reduce coefficients, but not eliminate them like the Lasso Regression model. This can have consequences on the results. A solution to this could be to potentially resort to using the Lasso Regression model and experimenting with the hyperparameters further to eliminate the coefficient and more to achieve more accurate results.

Conclusion

Overall, this research project aimed to explore the impact of existential factors on students' GPA, focusing on variables like weekly study time, parental support, absences, and tutoring. Using a ridge regression model in Python, we analyzed the data to assess the correlation between these factors and academic performance, evaluated by GPA. These model's effectiveness was measured through metrics like Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 to ensure proper results.

Our findings indicate that among the factors analyzed, absences had the most severe impact with a strong negative correlation with academic performance. Additionally, we were able to confirm that some activities/factors did not truly have effects on the students' performance.

In conclusion, this study contributes to the understanding of existential influences on educational environments, meaning that students have control over their academic performance and outcome.

Acknowledgments

The data used throughout this project was colleged by Rabie El Kharaoua and has been licensed by Attribution 4.0 International (CC BY 4.0). It can be accessed by using the following URL.
https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset/data

References

Bin Abdulrahman, Khalid A et al. "Study Habits of Highly Effective Medical Students." Advances in medical education and practice vol. 12 627-633. 8 June 2021, doi:10.2147/AMEP.S309535